

MESSOU FRANCK

AI Research Engineer · LLMs & Time-Series · Graph Neural Networks · Edge ML for IoT & Energy

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📍 Tokyo, Japan

in [mesabo](#)

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PhD candidate and ML engineer with 3+ years advancing LLM / small-language-model reprogramming for time-series forecasting and imputation, graph neural networks for power-system stability, and edge / IoT sensor intelligence. First-author of **2 Q1 journal papers** and **5 IEEE conference papers** (GLOBECOM, ICC, VTC, ICUFN), with the IEEE VTS 2024 Student Paper Award. Bridges research and production: LoRA / PEFT fine-tuning, distributed training (DDP), and vLLM serving on multi-GPU infrastructure, plus LangChain / LangGraph / RAG pipelines, alongside shipped AI products (CVFlow, AIReco). Available immediately for a **research engineer or applied-scientist role** in AI / ML.

KEY ACHIEVEMENTS

- First-author of **2 Q1 journals** (Internet of Things IF 7.6; Expert Systems with Applications IF 7.5) and **5 IEEE conference papers**; IEEE VTS 2024 Student Paper Award.
- Pioneered LLM / sLM reprogramming (Spec2Llama, Spec2LLM, LLM4Imp) for forecasting and imputation on edge, published at **IEEE ICCE / ICC 2026**.
- Designed dual-domain graph-neural-network architectures for power-system stability on the PEGASE **2,869-bus** grid.
- Provisioned and operate a multi-GPU server end-to-end (DDP, checkpointing, vLLM) shared across the research team.

EXPERIENCE

AI / ML Researcher (Doctoral)

YuLab, Hosei University

📅 Sep 2025 – Present

📍 Tokyo, Japan

- Built LLM / sLM reprogramming pipelines (Spec2Llama, Spec2LLM, LLM4Imp) adapting frozen LLMs and compact backbones (SmolLM, Phi, Qwen) for forecasting and imputation on edge, updating **under 9% of parameters** while matching or beating SAITS / TimeMixer baselines; published at IEEE ICCE / ICC 2026.
- Designed graph-constrained multi-LLM ensembles with structural causal models, lifting causal-discovery **F1 to 0.945** (+11 pts over PCMC1) and narrative completeness **5.6×** over a single-LLM baseline.
- Designed dual-domain graph-neural-network architectures for power-system stability on the PEGASE **2,869-bus** grid.
- Provision and operate the lab's multi-GPU server: distributed training (DDP), checkpointing, and vLLM inference serving in Docker, shared across the team.

ML Researcher (Master's)

YuLab, Hosei University

📅 2023 – Sep 2025

📍 Tokyo, Japan

- Built transformer time-series forecasting models (TAPformer, TSFormer, CNN-BiLSTM) for long-term electricity load forecasting, outperforming standard CNN / LSTM baselines (published in *Internet of Things*, Q1, IF 7.6).
- Won the IEEE VTS Tokyo / Japan 2024 Student Paper Award (VTC2024-Fall, Washington D.C.).

LLM Systems Engineer

EDUCATION

PhD, Applied Informatics

Hosei University

📅 2025 – 2027 (Expected)

📍 Tokyo

MSc, Applied Informatics

Hosei University

📅 2023 – 2025

📍 Tokyo

TECHNICAL SKILLS

Python PyTorch Hugging Face

Transformers LLM Fine-tuning

LangChain LangGraph LangSmith

RAG RAGAS FAISS Vertex AI

LoRA / PEFT Time-Series Forecasting

Imputation GNN Anomaly Detection

vLLM CUDA DDP Docker FastAPI

scikit-learn

SOFT SKILLS

Research Mindset Mentoring

Scientific Writing Autonomous

Cross-cultural Teamwork Rigorous

Curiosity Communication Adaptable

Proactive Collaborative

APPLIED WORK

Co-Founder & Lead

CVFlow ([cvflow.app](#))

📅 2025 – Present

📍 Remote

- Co-built a production AI resume platform; owned the LLM integration (Gemini / GPT) across 20 languages.

Open-Source / Personal Research Initiative

📅 2025 – Present 📍 Remote • [GitHub.com](#)

- Architected and deployed four open-source production LLM systems (semantic search, retrieval-augmented generation, stateful agent orchestration, full-stack LLMOps), shipped via **GitHub Actions CI/CD** to GCP (Cloud Run, GKE, Vertex AI).
- Engineered composable LCEL pipelines with hybrid retrieval (FAISS, ChromaDB, BM25 sparse, MMR re-ranking), evaluated with RAGAS (context recall, faithfulness, semantic similarity).
- Built stateful LangGraph agents (conditional routing, cycles, human-in-the-loop) and ReAct loops with Pydantic-validated tools; FastAPI streaming APIs with semantic caching and LangSmith tracing at sub-200ms TTFT.

LangChain LangGraph LangSmith FAISS ChromaDB
RAGAS GCP

Machine Learning Engineer

D'isum Inc.

📅 Oct 2025 – Jul 2026 📍 Tokyo, Japan — Hybrid

- Built ETL pipelines from raw sensor streams (50+ engines, 200+ batteries) and classical ML models for sound and time-series anomaly / fault detection.
- Built on-device iOS / Android sensor apps (Swift, Java) for real-time data capture and edge inference.

Generative AI Intern (Summer Internship)

日本AIコンサルティング (Japan AI Consulting)

📅 Summer 2024 📍 Osaka, Japan

- Customized AI model source code and integrated LLMs into interactive UIs using Gradio, delivering tailored outputs for internal use cases.
- Developed and maintained log data processing pipelines using FastAPI and Flask, documented via Growi.
- Produced research reports on data privacy and the AI industry landscape, supporting J-AIC's business expansion strategy.

RESEARCH PROJECTS

SmolSearch – Semantic Search API

FAISS • LCEL • FastAPI • Cloud Run • [Github.com](#)

- ★ LCEL chain (SmolLM2-360M) over a FAISS index; FastAPI on Cloud Run, sub-100ms TTFT, MMR search.

RAGify – Evaluated RAG System

ChromaDB • RAGAS • GCP • [Github.com](#)

- ★ Hybrid retrieval (ChromaDB, FAISS, MultiQuery, BM25) scored with RAGAS on GCP.

AgentFlow – Stateful Multi-Tool Agent

LangGraph • ReAct • LangSmith • GKE • [Github.com](#)

- ★ LangGraph StateGraph with Pydantic tools and full LangSmith tracing; deployed on GKE.

LLMOps Baseline – Production LLMOps Reference

LangChain • LangSmith • Vertex AI • FastAPI • [Github.com](#)

- ★ LCEL + Pydantic output, Redis caching, SSE streaming, tiktoken costs; GitHub Actions CI/CD to Vertex AI.

Spectral LLM / sLM Reprogramming (Spec2Llama • Spec2LLM •

LLM4Imp)

Frozen LLMs • PEFT • spectral prompts • PyTorch

Text in blue is clickable — it opens the linked profile, project, or paper.

Applied AI Engineer

AIReco

📅 2025 – 2026 📍 Remote

- Offline QLoRA-tuned sLM with cloud-LLM routing; ASR, summarization, and translation across 20 languages.

PARTICIPATION

- Represented J-AIC at JICA's TICAD9 booth, demoing applied ML / GenAI prototypes to international delegations, 2025.
- Presenter at IEEE international conferences (VTC, ICUFN, GLOBECOM, ICC).

AWARDS & CERTIFICATES

- IEEE VTS Tokyo / Japan Chapter **2024 Student Paper Award** & Encouragement Award (VTC2024-Fall).
- JICA Development Studies Program (JICA専用講座), 2025.
- Review Committee Member, **IEEE ICECET 2026** (IEEE Italy Section); Reviewer, *Theoretical Computer Science* (Elsevier).

SELECTED PUBLICATIONS

- TAPformer: Long-Term Electricity Load Forecasting in IoT Systems, *Internet of Things* (Elsevier), 2026 — Q1, IF 7.6.
- Spec2Llama: Spectral-Aware Forecasting for Versatile Time Series, *Expert Systems with Applications*, 2026 — Q1, IF 7.5.
- Hybrid Attention CNN-BiLSTM for Short-Term Load Forecasting, *IEEE VTC2024-Fall* — Student Paper Award.
- TSFormer: Temporal-Aware Transformer for Multi-Horizon Forecasting, *IEEE ICUFN 2025*.
- Spec2LLM / LLM4Imp: LLM reprogramming for forecasting and imputation, *IEEE ICCE / ICC 2026*.

LANGUAGES

French (native) ●●●●●

English (fluent) ●●●●●

Japanese (daily, ongoing) ●●●●●

- Adapt frozen language models for time-series forecasting and imputation on edge; published at IEEE ICCE / ICC 2026 and ESWA (Q1).

Transformer Forecasting for IoT (TAPformer · TSFormer ·

CNN-BiLSTM)

attention · multi-horizon · long-term load forecasting

- Transformer architectures for electricity-load forecasting in IoT, outperforming CNN / LSTM baselines; 2 Q1 journals and IEEE conference papers.

Multi-GPU MLOps Stack

DDP · vLLM · CUDA · Docker · Linux

- Provisioned and operate reusable training and inference-serving infrastructure for the lab.

ACADEMIA

Research Assistant (part-time)

YuLab, Hosei University

📅 2023 – Present

📍 Tokyo, Japan

- Assist faculty with data curation and processing, model-training environment setup, and paper drafting.

Teaching Assistant

Hosei University

📅 2023 – Present

📍 Tokyo, Japan

- Support English-medium classes (4th-year undergraduate and Master's graduate), helping communicate LLM and research concepts clearly in English.